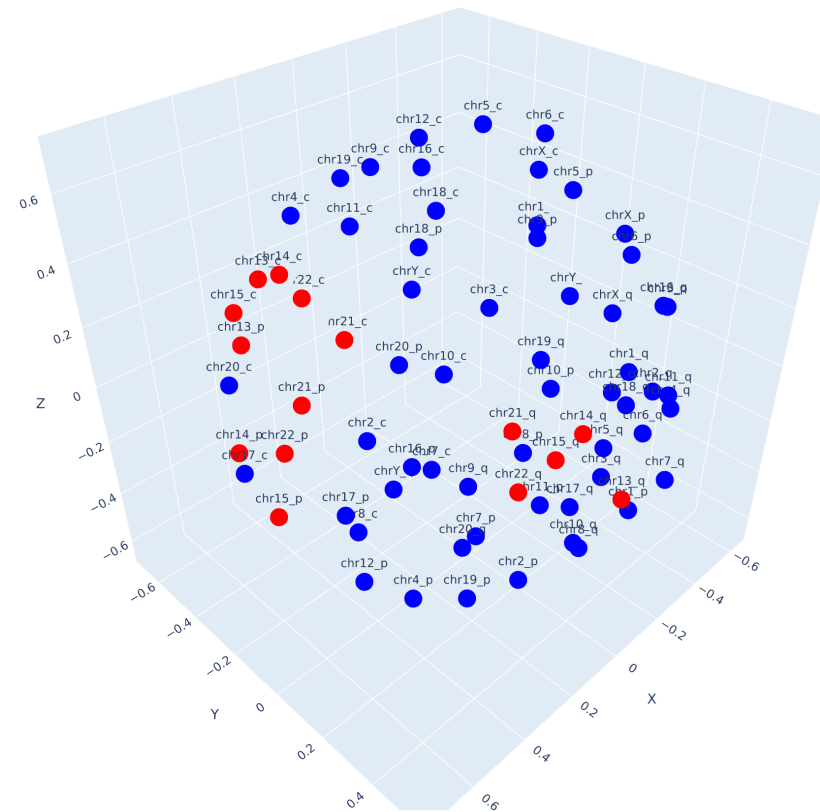


CHM13 Hi-C MDS is whole-arm contact space

3D Visualization of Chromosomal Data using MDS

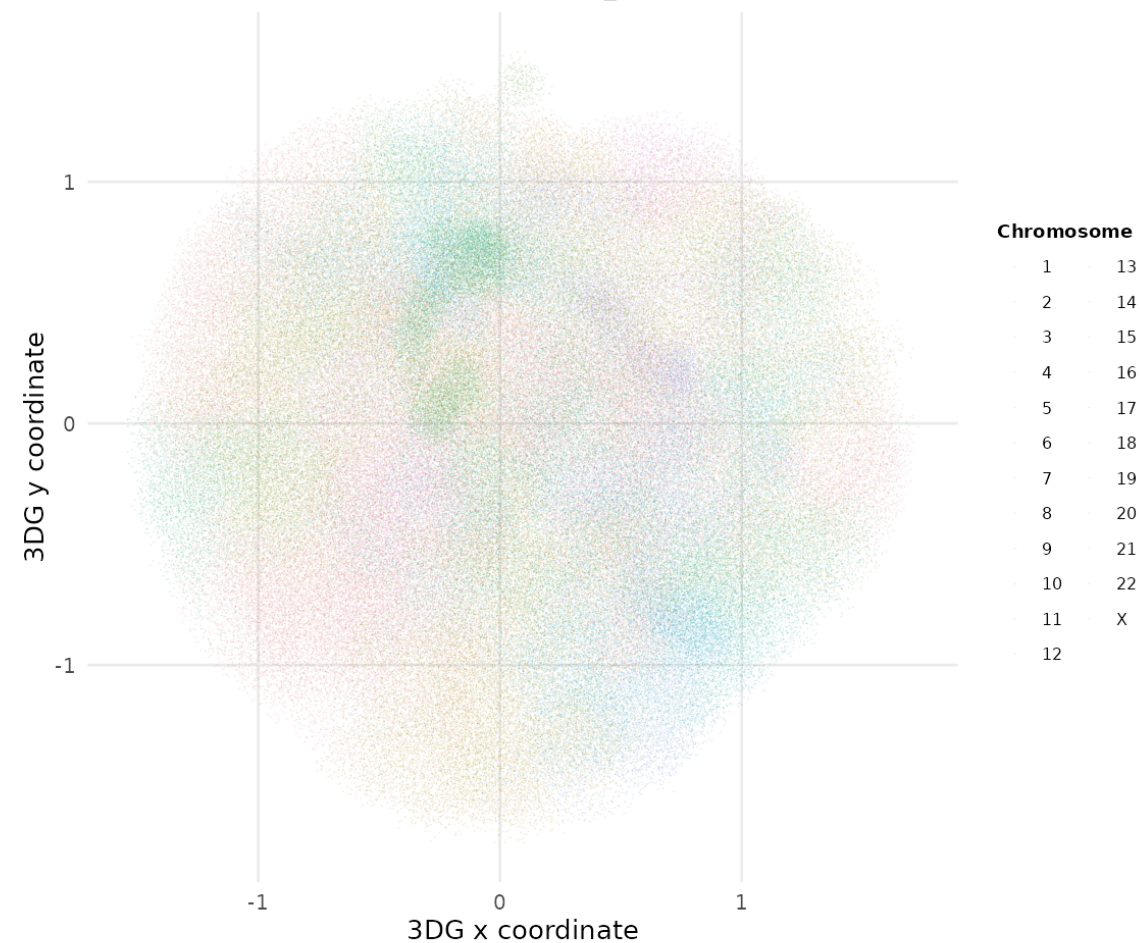


Region definition: whole CHM13 p-arm, centromere, and q-arm intervals from CHM13_chrom_parts.bed (72 regions). This is **not** PHR intervals, **not** terminal 500 kb, and **not** 1 Mb PHR flanks/windows. MDS embeds contact frequencies from a broader CHM13 .mcool arm/centromere matrix; it is a contact-space summary, not a physical single-cell genome reconstruction.

One existing GM12878 Dip-C cell gives a whole-genome 3DG view

GM12878 Dip-C whole-genome 3D coordinate projection

One existing reconstructed cell (gm12878_01); 2D view of x/y coordinates from 3DG particles

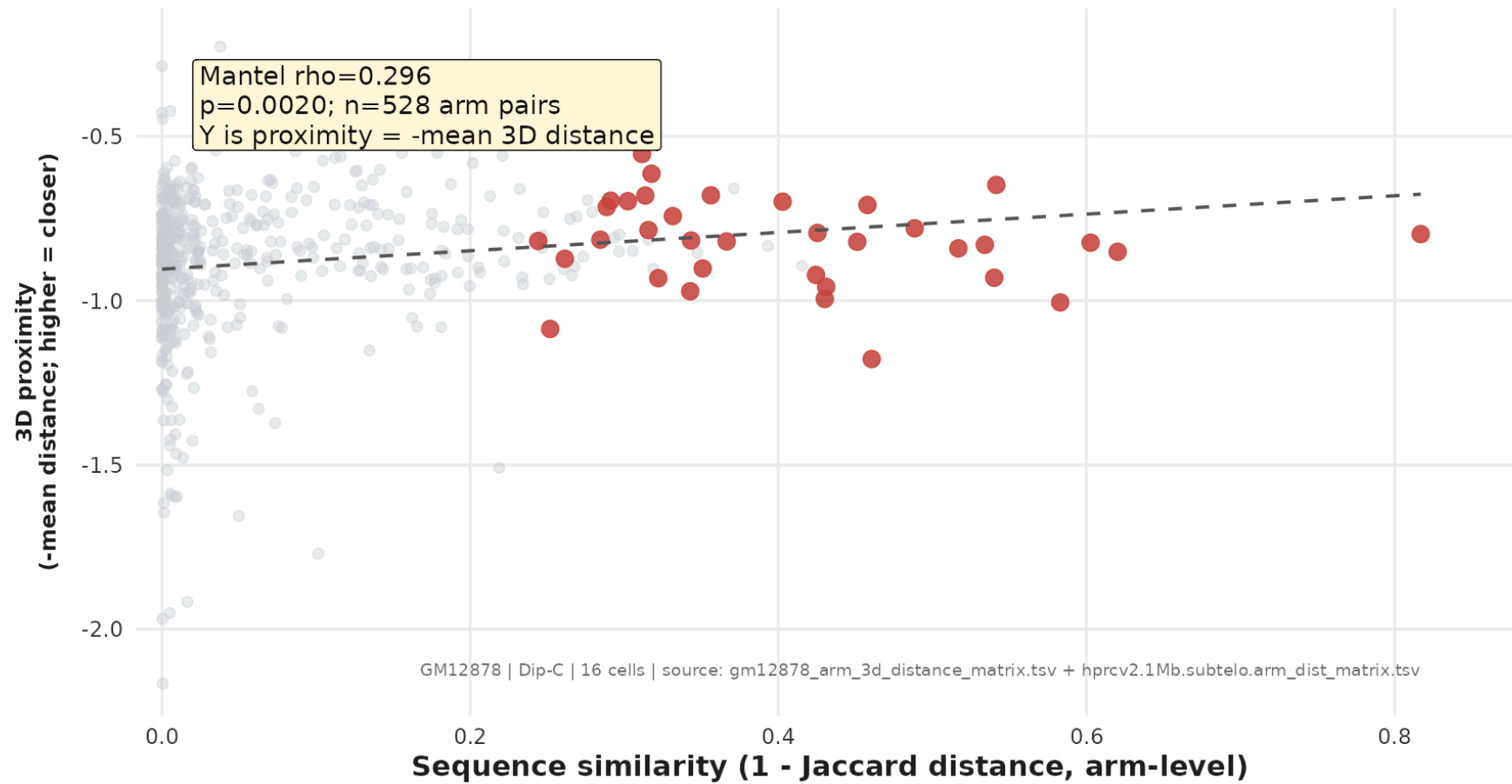


Optional candidate only: this is a 2D x/y projection of one existing 3DG file (gm12878_01) from single-cell Dip-C. It is physical-coordinate context, not CHM13 Hi-C and not a PHR/community validation plot.

GM12878 Dip-C: sequence similarity predicts 3D proximity

GM12878: sequence similarity vs 3D proximity

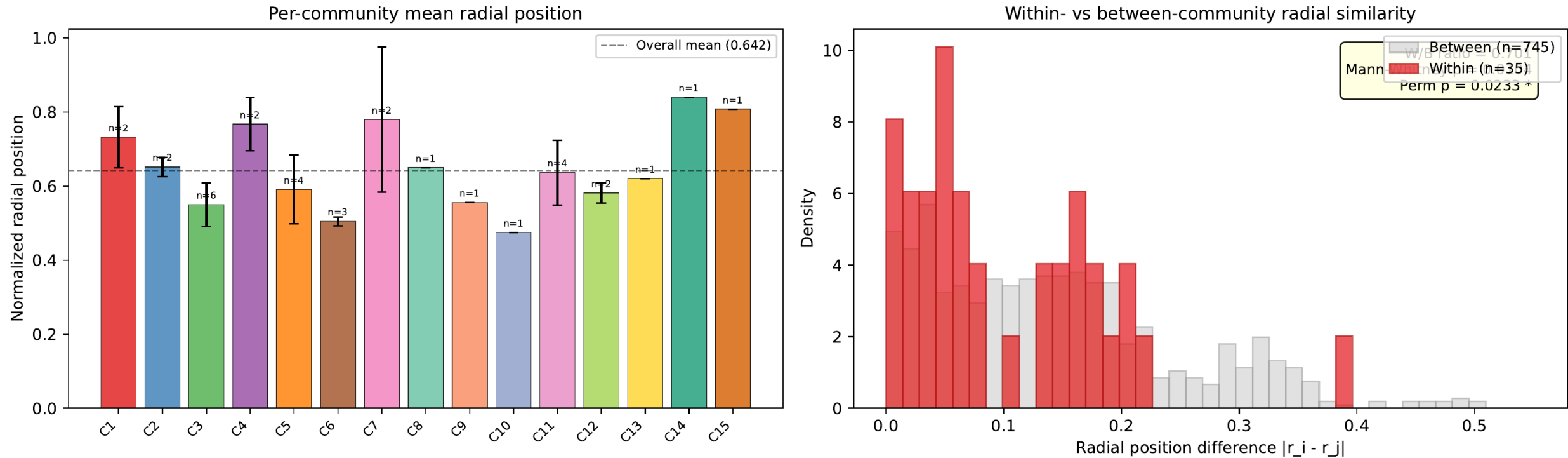
Higher X = more shared subtelomeric sequence; higher Y = closer in 3D



W/B means within-community divided by between-community 3D distance; W/B < 1 means within-community arms are closer. Proximity convention: 3D proximity = -mean 3D distance, so higher y-values are closer and a positive rho has the expected direction.

GM12878 Dip-C radial community structure

Radial proximity community analysis (Dip-C)



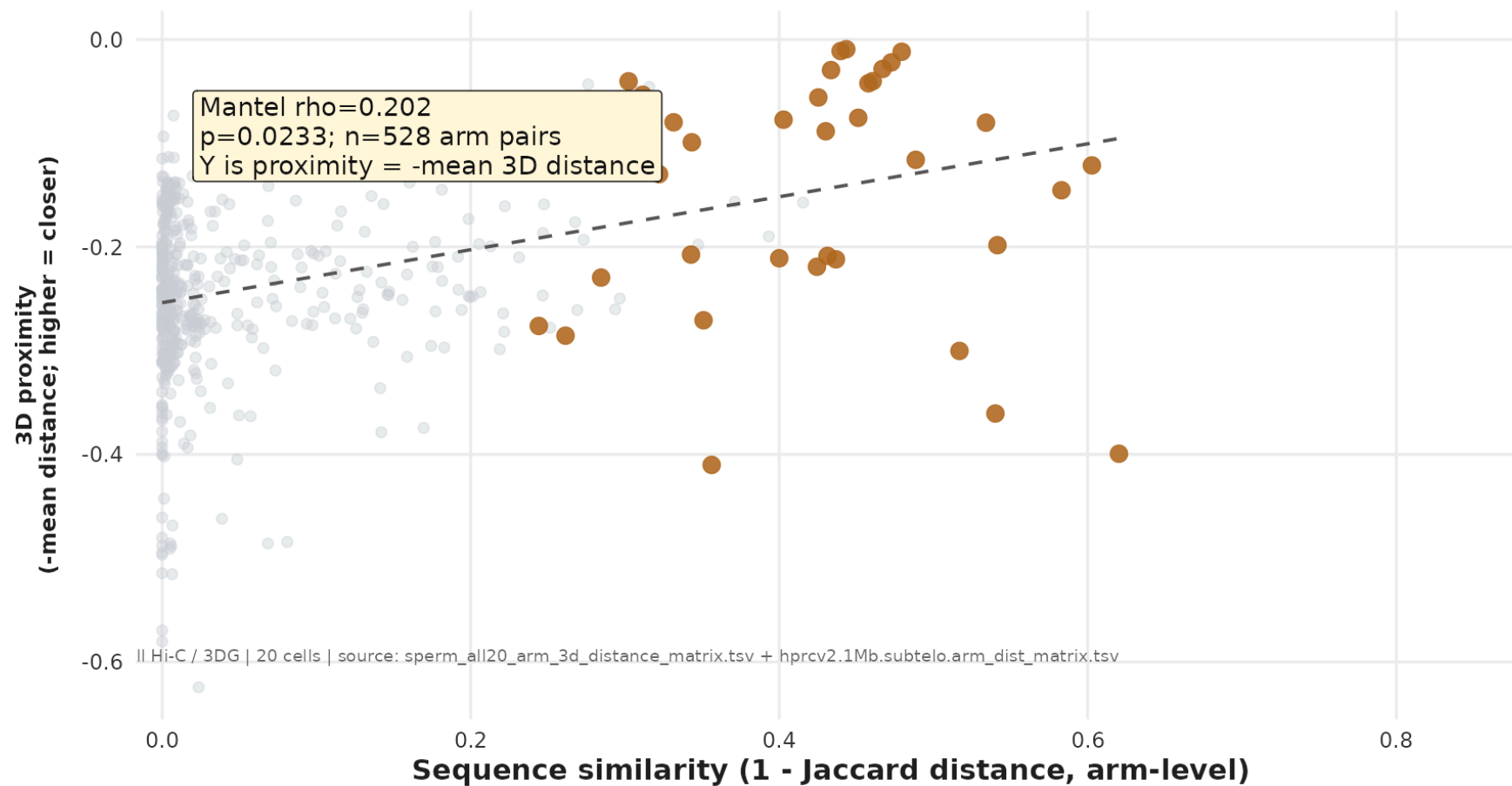
Radial position = distance from nuclear center (mean of all particles), normalized by 95th percentile radius. Panel A: per-community mean $\pm$ SD of arm radial positions. Panel B: pairwise $|r_i - r_j|$ for within- vs between-community arm pairs.

This radial panel is GM12878 only: per-community normalized radial position plus within-vs-between radial similarity. Same-community arms have more similar radial positions than between-community arms.

Sperm single-cell Hi-C: sequence similarity predicts 3D proximity

Sperm: sequence similarity vs 3D proximity

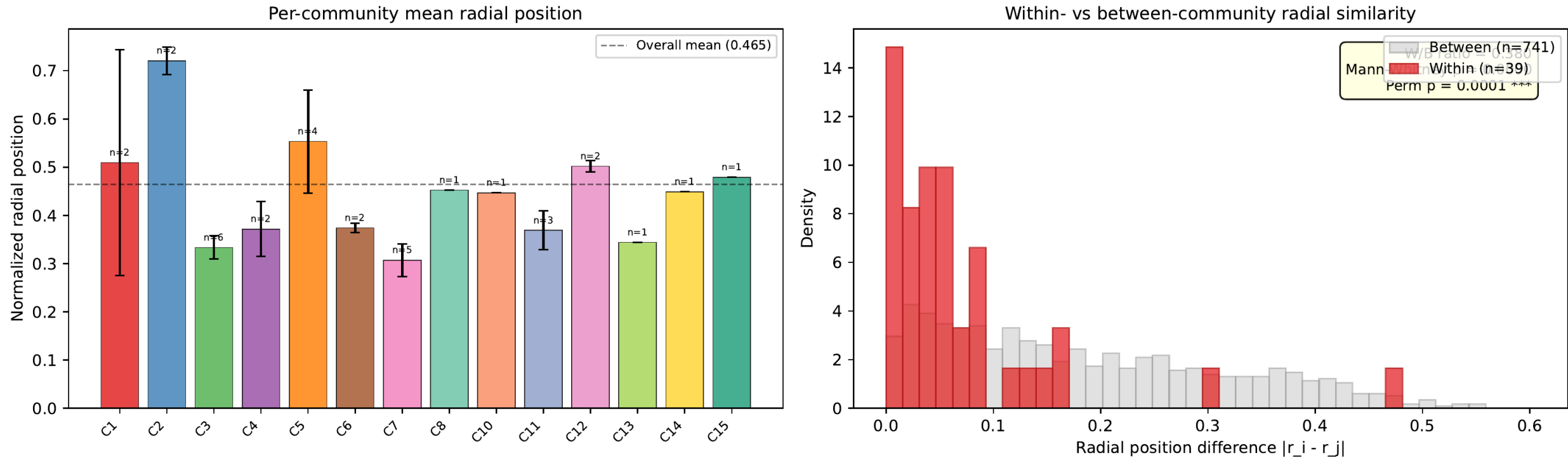
Higher X = more shared subtelomeric sequence; higher Y = closer in 3D



W/B means within-community divided by between-community 3D distance; W/B < 1 means within-community arms are closer. Proximity convention: 3D proximity = -mean 3D distance, so higher y-values are closer and a positive rho has the expected direction.

Sperm single-cell Hi-C radial community structure

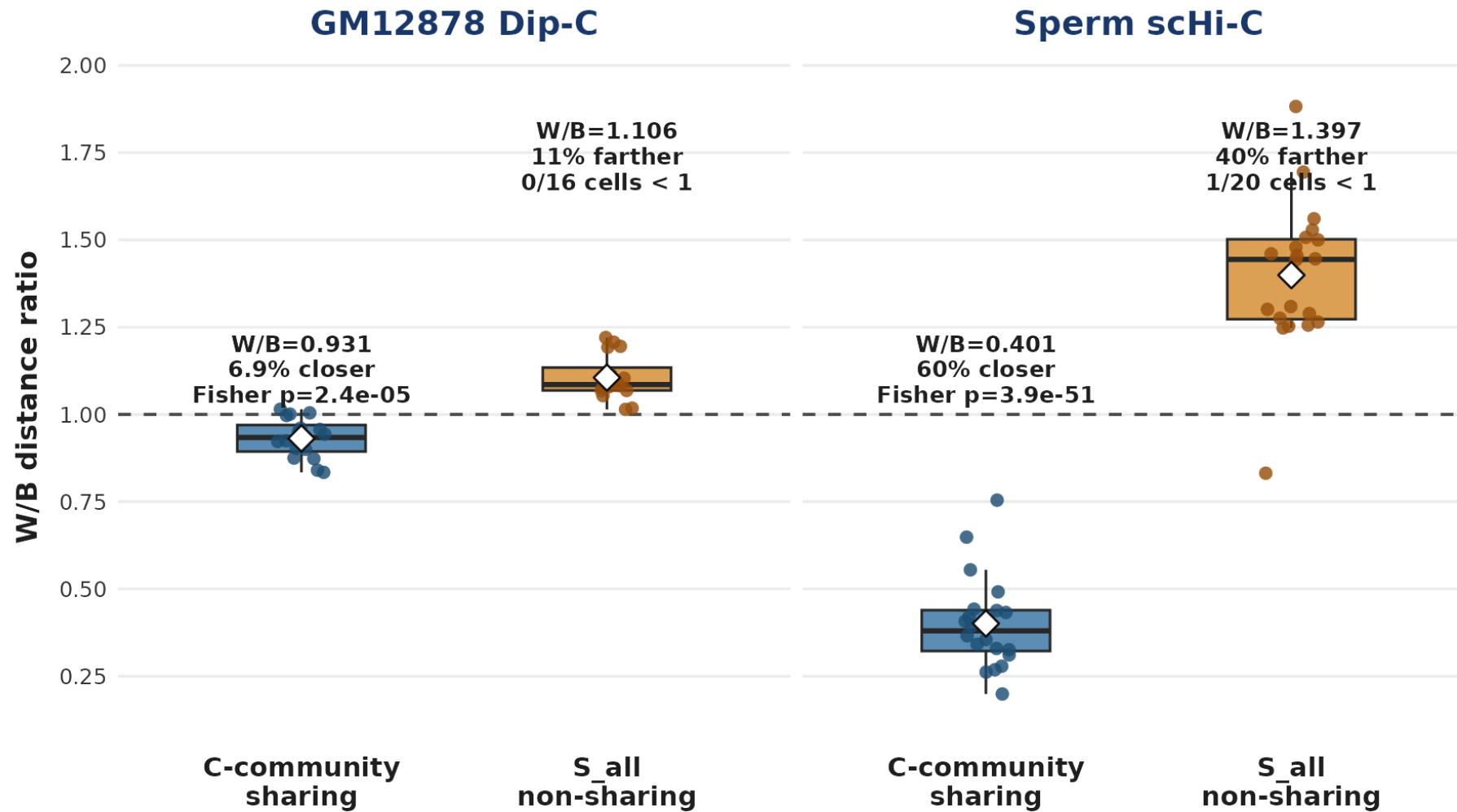
Radial proximity community analysis (Dip-C)



Radial position = distance from nuclear center (mean of all particles), normalized by 95th percentile radius. Panel A: per-community mean $\pm$ SD of arm radial positions. Panel B: pairwise $|r_i - r_j|$ for within- vs between-community arm pairs.

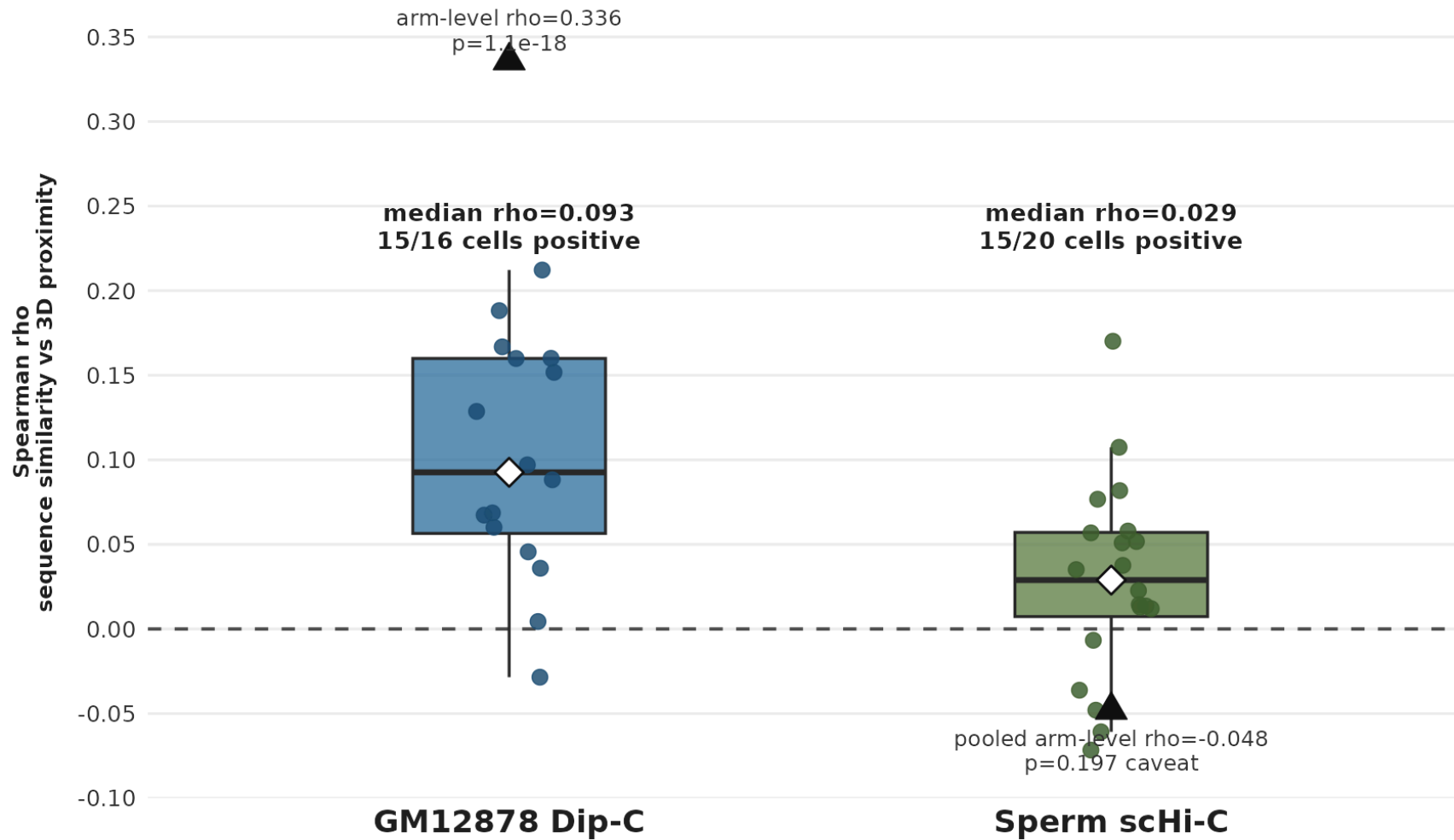
This radial panel is sperm only: per-community normalized radial position plus within-vs-between radial similarity across 20 haploid sperm cells. It measures the same radial statistic as the GM12878 radial panel, in a distinct cell type.

Negative control: non-sharing S_all arms are farther apart



Reduced text scale version of slide 11b. W/B is within-community divided by between-community 3D distance; values below 1 mean closer within-community, while S_all values above 1 show the non-sharing negative control is farther apart.

Community-free per-cell rho: sequence similarity predicts proximity



Reduced text scale version of slide 11c. Rho is computed as sequence similarity versus 3D proximity (-distance), so positive rho means more similar sequences are closer in 3D.